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BHARAT PROCESS INDUSTRIES PVT. LTD.
INCIDENT REPORT FORM

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Report No : IR-2024-007
Date of Report : 06-March-2024
Report Prepared By : Rajan Kumar (Technician Grade-II)
Reviewed By : Vikram Reddy (Maintenance Supervisor)
Approved By : Anil Gupta (Plant Manager)

SECTION 1 – INCIDENT DETAILS

Incident Date : 05-March-2024
Incident Time : 14:35 hrs (IST)
Location : Building C, Ground Floor, Unit 3
Equipment : Cooling Water Pump 104 (Tag: P-104)
Incident Type : Equipment Failure – Bearing Failure (Unplanned Shutdown)
Severity : HIGH – Production impact for 4.5 hours

SECTION 2 – WHAT HAPPENED

At 14:35 hrs on 05-March-2024, the DCS (Distributed Control System) triggered a high bearing temperature alarm on Pump P-104. The alarm setpoint is 75°C. Operator on duty (Mr. Prakash Tiwari) acknowledged the alarm and initiated field check.

By 14:42 hrs, field technician confirmed bearing temperature at 78°C and rising. Pump was manually tripped at 14:45 hrs per SOP-CW-003 (Section 4.2 – Emergency Shutdown Procedure).

Post-shutdown inspection at 15:10 hrs revealed Drive End (DE) bearing had failed. Physical inspection showed:

- Bearing cage fragmented (SKF 6309)
- Significant heat discolouration on bearing housing (blue/brown marks)
- Grease carbonised – indicating overheating prior to failure
- No evidence of water ingress or external contamination

Standby pump (P-105) was started at 15:20 hrs after system checks. Normal cooling water flow restored to Unit 3 at 15:35 hrs. Total production disruption: 4 hours 55 minutes.

SECTION 3 – ROOT CAUSE ANALYSIS

Immediate Cause:

Drive End bearing (SKF 6309) failed due to thermal overload caused by inadequate lubrication.

Contributing Factor 1 – LUBRICATION INTERVAL BREACH:

Previous lubrication record (WO-2024-001, dated 10-January-2024) shows last greasing was 54 days before this failure. OEM recommended interval (OEM-KBL-KDS450-001, Section 6.3) specifies maximum 45 days between lubrication cycles for this pump class. The PM schedule had NOT been updated to reflect OEM requirement – it was set at 90 days (quarterly).

Contributing Factor 2 – ELEVATED TEMPERATURE TREND IGNORED:

WO-2024-001 (10-Jan-2024) already noted bearing temperature at 72°C – close to the 75°C alarm limit. No follow-up action was scheduled and no alert was raised to the maintenance supervisor at that time.

Contributing Factor 3 – NO CONDITION MONITORING:

Pump P-104 has no online vibration monitoring. Vibration was last manually checked on 10-January-2024. If continuous monitoring was in

place, bearing degradation would have been detectable 7-10 days before failure via vibration signature changes.

Root Cause (Systemic):

PM schedule for Pump P-104 (and P-105) was not aligned with OEM specifications. No escalation mechanism existed for near-limit temperature readings noted in maintenance records.

SECTION 4 – IMMEDIATE CORRECTIVE ACTIONS TAKEN

1. Bearing replaced with SKF 6309 (same spec – OEM approved)
2. Bearing housing cleaned and inspected – no damage to shaft or housing
3. Fresh grease applied (SKF LGEP 2 – OEM specified grade)
4. Pump recommissioned and performance verified
5. Pump run for 2 hours under observation before handing over to operations

SECTION 5 – PREVENTIVE ACTIONS (Long Term)

- Action 1 : Revise PM schedule for P-104 and P-105 lubrication from 90 days to 30 days. (Owner: Suresh Nair | Due: 10-Mar-2024)
- Action 2 : Add rule in DCS – any bearing temperature reading above 70°C to trigger supervisor notification (not just operator alarm). (Owner: Instrumentation Dept – Rajesh Pillai | Due: 31-Mar-2024)
- Action 3 : Conduct audit of ALL rotating equipment PM schedules vs OEM manuals to identify similar gaps. (Owner: Vikram Reddy | Due: 30-Apr-2024)
- Action 4 : Evaluate feasibility of online vibration monitoring for P-104, P-105, and BL-07 blower. (Owner: Anil Gupta | Due: 31-May-2024)
- Action 5 : Add "near-limit temperature readings" to the list of items requiring supervisor sign-off in maintenance work orders. (Owner: Anil Gupta | Due: 15-Mar-2024)

SECTION 6 – REGULATORY COMPLIANCE NOTE

This incident has been reviewed against OISD-STD-137 (Inspection of Rotating Equipment). The failure is classified as a maintenance-related incident, not a process safety incident. No reportable event under Factory Act 1948 (Section 88A) as no injury or fire occurred. Incident to be recorded in plant incident log as required under Factory Act Rule 103 (Maharashtra Factory Rules).

SECTION 7 – SIMILAR INCIDENTS – HISTORICAL CHECK

- A search of incident records was conducted:
- IR-2022-014 (Sep 2022): P-105 mechanical seal failure – different root cause (dry running). Not directly related.
 - IR-2021-003 (Feb 2021): P-104 bearing failure – SAME pump, same bearing. Review of IR-2021-003 shows same contributing factor: lubrication interval was 67 days at time of that failure. CORRECTIVE ACTION FROM IR-2021-003 WAS NOT SUSTAINED – PM schedule was apparently reset to 90 days at some point after 2021 without documented justification.

NOTE: The recurrence of same failure on same equipment is of concern. This must be escalated to the Engineering Manager for review.

Signatures:

Prepared by : Rajan Kumar	Date: 06-Mar-2024
Reviewed by : Vikram Reddy	Date: 07-Mar-2024
Approved by : Anil Gupta	Date: 08-Mar-2024